

BRIAN MAK

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<https://bmak.me/> · <https://linkedin.com/in/brian-mak> · <https://github.com/eggroll-bot>

SKILLS

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|---------|----------|----------------|-------|
| • C/C++ | • Python | • Bash | • x86 |
| • Lua | • Git | • Linux kernel | • ARM |

EXPERIENCE

SOFTWARE ENGINEER, LINUX OS / KERNEL

HEWLETT PACKARD ENTERPRISE, FORMERLY JUNIPER NETWORKS

L3, JUL 2025 – CURRENT | L2, JUL 2024 – JUL 2025 | L1, APR 2024 – JUL 2024 | INTERN, JUN 2023 – SEP 2023

- Played a key role in the first ARM platform bring-up for Junos OS Evolved, resolving complex bootloader, Linux kernel, and userspace issues and integrating an ARM vendor kernel with Juniper's x86 kernel to unify both architectures under a single kernel, enabling deployment of Junos OS Evolved on ARM SoCs with a single maintainable codebase
- Led a major Linux kernel upgrade, including patch rebasing, driver refactoring, compiler-warning elimination, and debugging complex kernel and open-source userspace regressions, enabling adoption of newer kernel features and improvements while improving the security posture through cleaner, warning-free code
- Delivered major kernel footprint and boot-time optimizations, reducing crashkernel reserved memory by 70%, optimizing kernel code size, enabling Junos OS Evolved to run on low-memory ARM SoCs, and reducing boot time by ~25 seconds on low-end platforms
- Enhanced application core dump and vmcore debugging infrastructure, including support for diagnosing truncated core dumps, enabling faster root-cause analysis and resolution of customer issues involving crashes
- Added x86 device tree loading support to the kexec_file_load syscall, enabling the transition from the legacy kexec_load syscall while preserving device tree functionality and strengthening the security posture through kernel integrity validation with IMA and secure boot
- Root-caused and resolved complex Linux kernel and userspace issues, including undefined behavior, resulting in multiple customer and internal wins across the following areas:
 - Kernel: I2C/SMBus, device tree, networking, platform/chipset drivers, IMA, TPM, RNG, watchdog
 - Userspace: systemd, FUSE filesystems, util-linux, kexec-tools, vendor SDKs

EMBEDDED CAPTURE THE FLAG COMPETITION

MITRE, JAN 2023 – APR 2023

- Participated on a team for a cybersecurity CTF competition to design and implement a secure key fob system on an ARM microcontroller for a car door lock using Rust, and attack opposing teams' designs.
- Designed and implemented 4+ drivers/modules to support the operation of the key fob and car firmware, including the EEPROM, random number generation, GPIO button, and timer modules.
- Designed and implemented the secure car unlocking and key fob pairing transactions.

SOFTWARE ENGINEERING INTERN, PLATFORMS (TIMING)

JUNIPER NETWORKS, JUN 2022 – SEP 2022

- Designed and implemented a history library to track IEEE 1588 Precision Time Protocol state changes, allowing engineers and testers to troubleshoot errors and debug code more efficiently.
- Implemented software debug features for an FPGA implementing IEEE 1588 Precision Time Protocol clocks, allowing engineers and testers to effectively debug packets discarded by the FPGA.

EDUCATION

COMPUTER SCIENCE, B.S., TECHNOLOGY AND INFORMATION MANAGEMENT MINOR

UNIVERSITY OF CALIFORNIA, SANTA CRUZ

- GPA: 3.94, 8x Dean's Honors List